Romesh Prasad

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Research Profile

My research develops intelligent decision-making methods for manufacturing and operations systems, with emphasis on reinforcement learning, combinatorial optimization, multi-agent coordination, and deployable AI. I study how algorithms can sense changing operating conditions, make resource and scheduling decisions, and translate those decisions into coordinated physical action. My work spans modeling, algorithm development, large-scale GPU computing, and validation on physical systems.

Academic Appointments

Postdoctoral Research Fellow, University of Rhode Island — Feb 2024 – Present

- Researcher on a **$10M Office of Naval Research program** developing resilient, AI-enabled manufacturing and operations systems.
- Formulated the **Vehicle Routing Problem with Resource-Constrained Pickup and Delivery (VRP-RPD)**, a new combinatorial optimization problem for coordinating transportation and redeployable resources; developed a problem-specific BRKGA accepted at GECCO 2026 and nominated for **Best Paper**.
- Develop reinforcement learning and multi-agent reinforcement learning approaches for large-scale routing and resource coordination, including systematic comparisons with metaheuristic optimization.
- Contribute to **BrkgaCuda 3.0**, a multi-GPU asynchronous island model for large-scale biased random-key genetic algorithms, investigating scalable parallel search and information exchange across GPU populations.
- Developed a **vendor-agnostic coordination architecture** for heterogeneous industrial equipment and validated it across material-handling and manufacturing testbeds containing robots, AGVs, conveyors, storage systems, CNC equipment, and hardware from multiple vendors.
- Developed AI-enabled decision systems including multimodal deep learning for CNC monitoring, vision-language-model-guided dynamic routing, and an agentic LLM pipeline for translating natural-language optimization problems into mathematical formulations and solver-ready inputs.

Deep Reinforcement Learning Research Intern, National Renewable Energy Laboratory (DOE) — May 2023 – Aug 2023

- Developed an attacker-defender reinforcement learning framework to discover failure modes in power-grid frequency control systems.
- Benchmarked PPO, A2C, DQN, DDPG, and Q-learning and demonstrated that learned agents could identify and concentrate attacks on high-value network components.
- Parallelized RL experiments across **120 attack scenarios** using high-performance computing and automated experiment workflows.
- Published results at the **IEEE Power & Energy Society General Meeting 2024** and contributed to a DOE LDRD research proposal.

Doctoral Researcher, Syracuse University — Aug 2017 – May 2024

- Developed reinforcement-learning methods for autonomous recovery of cyber-manufacturing systems following sensor failures and false-data injection attacks.
- Designed a two-stage recovery framework for rerouting in-process and newly arriving jobs around unavailable resources while maintaining ongoing production.
- Deployed RL policies on a physical manufacturing testbed with **4 robots, 1 AGV, and 3 manufacturing processes**, reducing recovery downtime by **12%**.
- Developed a cyber-physical failure classifier combining network and physical-system alerts, achieving an **89% F1 score**, and investigated LSTM-based state forecasting for system recovery.
- Published **seven peer-reviewed papers** from this research across manufacturing, automation, cyber-physical systems, and engineering venues.

Education

Syracuse University, Ph.D., Mechanical Engineering — Aug 2017 – May 2024

- **Dissertation:** Recovery of Cyber Manufacturing Systems from False Data Injection Attacks on Sensors by Reinforcement Learning
- **Relevant Coursework:** Deep Reinforcement Learning, AI for Manufacturing, Stochastic Processes, Modeling &

Simulation

Syracuse University, M.S., Mechanical Engineering

University of Mumbai, B.Tech., Mechanical Engineering June 2012 – June 2016

Publications

*Full publication record and citation counts available through Google Scholar. *Corresponding author.*

Journal Articles

- Eacuello, S.S., **Prasad, R.S.**, Sodhi, M.S. "Multimodal Sensor Fusion and Temporal Deep Learning for Computer Numerical Control Toolpath and Condition Classification: A Cross-Validated Ablation Study." *Sensors*, 26(8), 2405, 2026.
- **Prasad, R.**, Moon, Y. "Recovery of Cyber Manufacturing Systems from False Data Injection Attacks on Sensors Using Reinforcement Learning." *Advanced Manufacturing*, 2(1), 1–24, 2025.
- **Prasad, R.**, Mehr, S.A.Z., Moon, Y. "Recovery Systems Architecture for Cyber-Manufacturing Systems Against Cyber-Manufacturing Attacks: Reinforcement Learning Approach." *Manufacturing Letters*, 35, 851–860, 2023.
- **Prasad, R.**, Moon, Y. "Architecture for Preventing and Detecting Cyber Attacks in Cyber-Manufacturing System." *IFAC-PapersOnLine*, 55(10), 2246–2251, 2022.
- Zhang, S., Shapiro, N., Gehrke, G., Castner, J., Liu, Z., Guo, B., **Prasad, R.**, et al. "Smartphone App for Residential Testing of Formaldehyde (SmART-Form)." *Building and Environment*, 148, 567–578, 2019.

Conference Papers

- Sodhi, M., **Prasad, R.** "Vehicle Routing Problem with Resource-Constrained Pickup and Delivery: A Heuristic-Informed BRKGA with Pattern-Based Analysis." *Proceedings of the Genetic and Evolutionary Computation Conference (GECCO)*, 188–196, 2026. **Best Paper Nominee.**
- **Prasad, R.**, Hassanaly, M., Zhang, X., Sahu, A. "Discovery of False Data Injection Schemes on Frequency Controllers with Reinforcement Learning." *IEEE Power & Energy Society General Meeting*, 2024.
- **Prasad, R.**, Moon, Y. "Comprehensive Analysis of Cyber-Manufacturing Attacks Using a Cyber-Manufacturing Testbed." *ASME International Mechanical Engineering Congress and Exposition*, 2022.
- **Prasad, R.**, Swanson, M.K., Moon, Y. "Recovering From Cyber-Manufacturing Attacks by Reinforcement Learning." *ASME International Mechanical Engineering Congress and Exposition*, 2022.
- **Prasad, R.**, Moon, Y. "Adaptive Intrusion Detection System for Cyber-Manufacturing System." *ASME International Mechanical Engineering Congress and Exposition*, 2021.
- Akowuah, F., **Prasad, R.**, Espinoza, C.O., Kong, F. "Recovery-by-Learning: Restoring Autonomous Cyber-Physical Systems from Sensor Attacks." *IEEE International Conference on Embedded and Real-Time Computing Systems and Applications*, 61–66, 2021.

Manuscripts Under Review

- Saseendran, H., **Prasad, R.**, Sodhi, M. "BrkgaCuda 3.0: A Multi-GPU Island Model Framework for Biased Random-Key Genetic Algorithms." *Journal of Parallel and Distributed Computing*. **Under review.**
- **Prasad, R.**, Sodhi, M. "Plug-and-Plan: A Minimal Coordination Interface for Vendor-Agnostic Orchestration of Heterogeneous Industrial Systems." *IEEE Transactions on Automation Science and Engineering*. **Under review.**
- **Prasad, R.**, Sodhi, M.. "Redeployable Resources in AGV-Integrated Job Shops: A Correspondence with Vehicle Routing." *International Journal of Production Research*. **Under review.**
- Sodhi, M. **Prasad, R.**, Saseendran, H. "Tunable Recombination in Resource-Constrained Pickup and Delivery: A Two-Block Random-Key Representation, Injection Operators, and Certified Search for the VRP-RPD" *ACM Transactions on Evolutionary Learning and Optimization*. **Under review.**

Teaching Experience

Instructor, Reinforcement Learning for Combinatorial Optimization, University of Rhode Island Summer 2026

- Designed and taught an internal seminar series introducing reinforcement learning methods for combinatorial optimization.

Instructor of Record, Machine Learning for Industry 4.0, University of Rhode Island Aug 2025 – Dec 2025

- Designed and delivered an asynchronous machine-learning course, including syllabus, lectures, assignments, and JupyterHub-based coding exercises.
- Led the development and classroom pilot of a **six-level AI-use disclosure framework** for transparent use of

generative AI in student work; evaluated with 22 students and submitted findings and recommendations for broader university consideration.

Teaching Assistant, Data Analysis for Engineers, Syracuse University Fall 2021 – Fall 2023

- Supported a 100+ student undergraduate course across three offerings; delivered lectures on linear regression and decision-tree modeling and developed applied data-analysis exercises.

Teaching Assistant / Lab Instructor, Introduction to SolidWorks, Syracuse University Spring 2021 – Spring 2023

- Led four laboratory sections per week across three course offerings, providing hands-on instruction in engineering design and CAD.

Additional Research and Industry Experience

Graduate Researcher, Syracuse University May 2018 – May 2019

- Investigated formaldehyde emissions from residential materials using controlled experiments, HPLC validation, CFD modeling, and building-energy simulation; contributed to the SmART-Form smartphone application and a publication in *Building and Environment*.

Process Improvement Intern, Byrne Dairy Jan 2019 – May 2019

- Applied statistical analysis, design of experiments, Six Sigma, and process monitoring to a multimillion-dollar manufacturing improvement project, increasing process efficiency by **23%**, reducing downtime by **15%**, and contributing to approximately **$700K/year** in additional revenue.

Geographic Information Systems Analyst, HERE Maps Jan 2017 – July 2017

- Analyzed LiDAR-derived road geometry for autonomous-vehicle mapping and contributed to a workflow improvement that reduced processing downtime by **20%**.

Honors and Awards

Best Paper Nominee, Genetic and Evolutionary Computation Conference (GECCO)	2026
Best Poster and Research Award, Syracuse University Research Day	2023
Invited Talk and Presentation, ASME IMECE	2022
Summer Research Fellowship, Syracuse University	2021, 2022
30% Scholarship, Syracuse University Master's Education	2022

Professional Service

Reviewer, ASME International Mechanical Engineering Congress and Exposition (IMECE) 2020 – Present

Technical Expertise

Reinforcement Learning: DQN, PPO, SAC, A2C, DDPG, multi-agent RL, environment and reward design
Optimization: Linear and mixed-integer programming, stochastic models, vehicle routing, scheduling, BRKGA, genetic algorithms, branch and bound
Machine Learning: MLP, RNN, LSTM, Transformers, vision-language models, graph neural networks, multimodal learning
High-Performance Computing: CUDA, GPU programming, multi-GPU optimization, parallel computing
Software and Frameworks: Python, PyTorch, Ray RLlib, Stable-Baselines3, Gymnasium, PyTorch Geometric, Gurobi, CPLEX, Hexaly, ROS2, Git